Intelligent Platform for Crowdsourcing (IPCrowd) Technology Preparations

Georgia Institute of Technology
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Purpose

Introduce/revisit some commonly used technology tools:

- Python
- Google Platforms – Google Colaboratory (Colab)
  - https://colab.google/
- Repository and Git version control
  - GT GitHub
    - https://drupal.gatech.edu/handbook/github-georgia-tech
    - https://support.cc.gatech.edu/support-tools/faq/what-gt-github-enterprise
  - GT GitLab
- IDE – VSCode
  - https://code.visualstudio.com/
- Figma
  - https://www.figma.com/
- Past Presentations
  - Students
  - Speakers
  - https://gatech.instructure.com/courses/341956/files/
- **NLP**
  - Introduction to NLP Presentation on Canvas
  - **Large Language Models (LLMs) – Hugging Face**
- **Flask**
  - **A simple app with LLM**

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Large Language Models – Hugging Face

Overview:

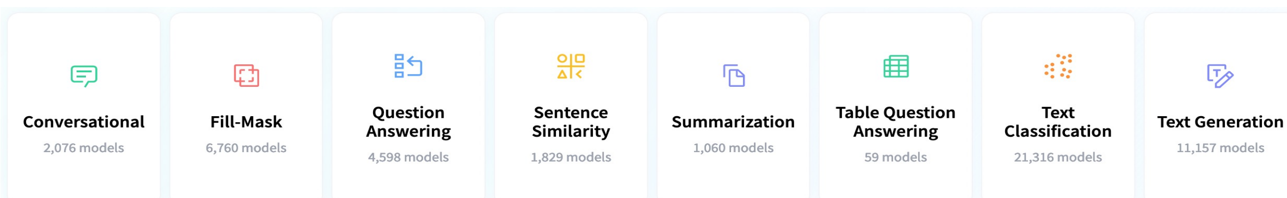
- Introduce Hugging Face
- Hugging Face Transformer Models
- Installation and Setup
- Examples to Run
- Potential Models to Use
- References

Hugging Face is a repository containing large amount of open source Machine Learning models for fields including:

- computer vision
- natural language processing (NLP)
- audio
- multimodal
- reinforcement learning

In NLP Field:

- Support multiple languages
- Models include
  - Bidirectional Encoder Representations from Transformers (BERT) [1]
  - Denoising autoencoder for pretraining sequence-to-sequence model (BART) [2]
  - GPT-2 [3]
  - Lots more



Hugging Face Transformer Models

- Transformers
  - Hugging Face NLP models trained on large data
- Text Classification Example:
  - https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment
  - https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest
- Other Tasks:
  - Question answering
  - Zero-shot classification
  - Summarization
  - Text classification
- Quick Tour
  - https://huggingface.co/docs/transformers/quicktour
  - Choose either Pytorch, TensorFlow, Flax

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Installation and Examples

Options (pick one):

1. Installation page: https://huggingface.co/docs/transformers/installation
2. Follow the `README.md`

Example code on Canvas `transformers_code`:

- Question answering task: `bart_question_answer.py`
- Zero shot classification task: `bart_zero_shot_classify.py`

bart_question_answer.py 2 X

bart_question_answer.py > ...

```python
"""
The script runs BART model on question answering task given a question and the answer in the text.
The model used could be found on:
    https://huggingface.co/docs/transformers/model_doc/bart#transformers.BartForQuestionAnswering.forward.example

"""

from transformers import AutoTokenizer, BartForQuestionAnswering
import torch

tokenizer = AutoTokenizer.from_pretrained("valhalla/bart-large-finetuned-squadv1")
model = BartForQuestionAnswering.from_pretrained("valhalla/bart-large-finetuned-squadv1")
question, text = "Who was Jim Henson?", "Jim Henson was a nice puppet"
#question, text = "What's the largest animal in the world?", "Elephant, ancient, friendly and largest mammal in the world."
inputs = tokenizer(question, text, return_tensors="pt")

with torch.no_grad():
    outputs = model(**inputs)

answer_start_index = outputs.start_logits.argmax()
answer_end_index = outputs.end_logits.argmax()
predict_answer_tokens = inputs.input_ids[0, answer_start_index : answer_end_index + 1]
print(tokenizer.decode(predict_answer_tokens, skip_special_tokens=True)) # print " nice puppet" or "Elephant" for the 2nd question and text

print(answer_start_index) # tensor(14)
print(answer_end_index) # tensor(15)
```

PROBLEMS 5   OUTPUT   DEBUG CONSOLE   TERMINAL   zsh

```
thuang38@ipsec-10-2-96-188 transformers_code % conda activate hugging-transformers
(hugging-transformers) thuang38@ipsec-10-2-96-188 transformers_code % python bart_question_answer.py
You passed along `num_labels=3` with an incompatible id to label map: {'0': 'LABEL_0', '1': 'LABEL_1'}. The number of labels wil be overwritten to 2.
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 nice puppet
tensor(14)
tensor(15)
(hugging-transformers) thuang38@ipsec-10-2-96-188 transformers_code % python bart_zero_shot_classify.py
Downloading model.safetensors: 100%|████████████████████████████████████████| 1.63G/1.63G [00:59<00:00, 27.3MB/s]
{'sequence': 'one day I will see the world', 'labels': ['travel', 'dancing', 'cooking'], 'scores': [0.9938650727272034, 0.003273803973570466, 0.00286103412
50896454]}
```

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Other Potential Models

- Sentiment analysis:
  - Sentiment analysis task: https://huggingface.co/models?pipeline_tag=text-classification&sort=downloads
- Group similar contents or posts:
  - Zero shot classification task: https://huggingface.co/models?pipeline_tag=zero-shot-classification&sort=downloads
  - Sentence similarity task: https://huggingface.co/models?pipeline_tag=sentence-similarity&sort=downloads
- Multiple models for each task available
- Example scripts available

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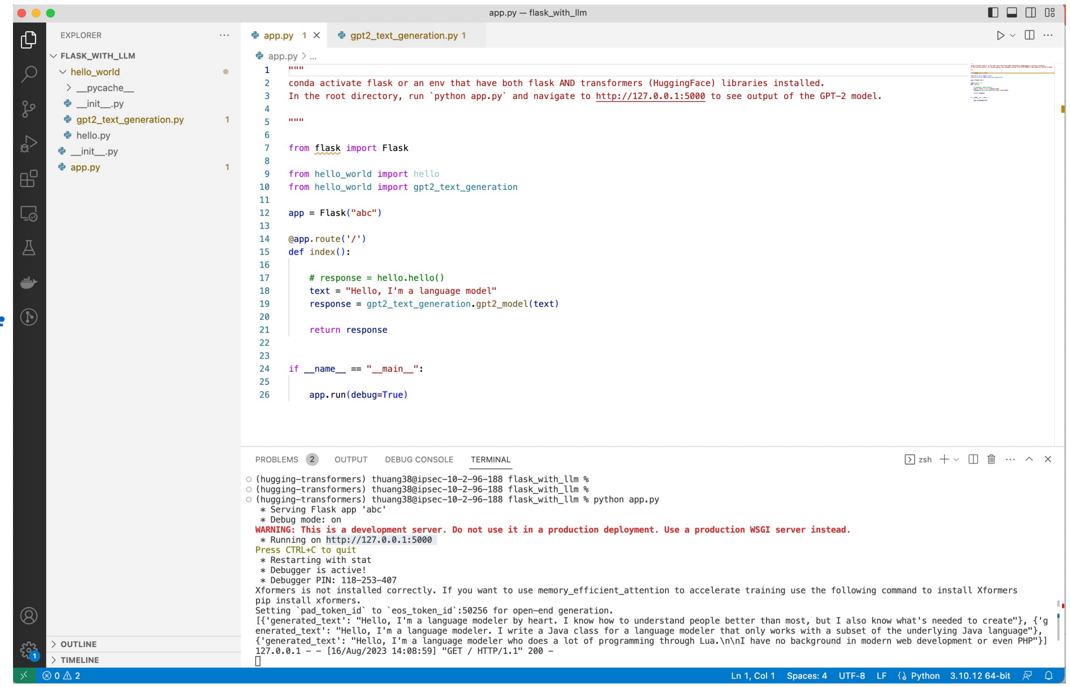
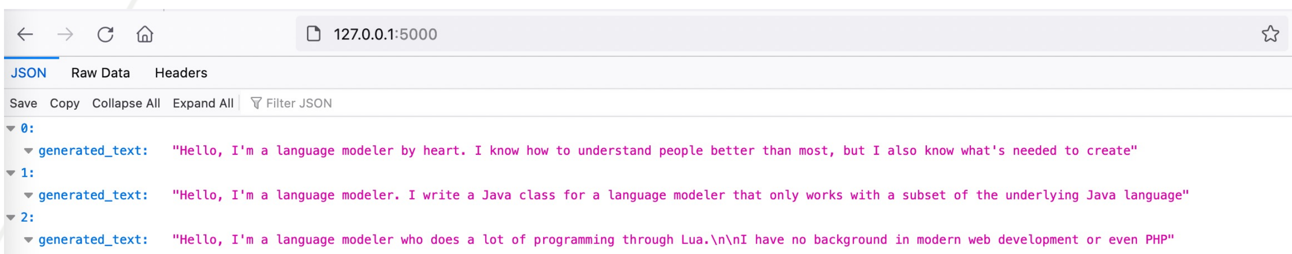
Flask with Large Language Model

Flask: A web framework in Python allowing development of web applications

https://flask.palletsprojects.com/en/2.3.x/installation/

Example code on Canvas: `flask_with_llm`





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References

1. Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
2. Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880, Online. Association for Computational Linguistics.
3. Radford, A., Wu, J., Child, R., Luan, D., Amodei, D. & Sutskever, I. (2018). Language Models are Unsupervised Multitask Learners. , .

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